Case 03 · Specification: Removal Sample Audit

Following chapter 2 (specification) of the Magshimim project-book template: promise, features by layer, decisions, what changed since the POC, and what stays open. Characters are fictional.

Opening scenario

Two camps cite the same number, 130 million, and conclude opposite things. One says most were dead and duplicates; the other says votes were stolen. Nobody knows how many deletions were wrong. The audit replaces the argument with a measurement: in each district, 1,500 randomly drawn deletions are checked in the field, and the result is one rate with a margin, a method and a seed anyone can rerun.

What the product promises

- For each district: the share of wrong deletions, with a 95% interval.
- A draw nobody can steer, and a result anyone can reproduce.
- Districts above a threshold get a full re-check before the final roll.
- What it does not promise: it is only as good as whoever knocks on the door.

Structure: four layers, ten features

Layer	Features
The draw	1. Freeze the deletion list · 2. Public draw · 3. Stratified sampler
In the field	4. Offline field app · 5. Revisit protocol · 6. Checker assignment
The estimate	7. Corrected estimate · 8. Booth flags
The report	9. District report and threshold · 10. Reproduction pack

Features

- **1. Freeze:** the list is locked and its SHA-256 published before anyone knows who will be drawn.
- **2. Public draw:** the seed comes from a public randomness beacon (NIST Beacon, drand), announced in advance.
- **3. Stratified sampler:** Neyman allocation; more visits to the opaque "absent, shifted or other" bucket, where the errors are.
- **4. Offline field app:** a PWA with IndexedDB and WebAuthn; every finding is signed, with time and place, even without coverage.
- **5. Revisit protocol:** a closed door gets two more visits; only then "unknown", never "shifted".
- **6. Checker assignment:** the checker must be independent of the deletion. Where thousands of such checkers come from is an open question.
- **7. Corrected estimate:** wrong-deletion rate, corrected for checker error via double visits, with a 95% interval (SciPy).
- **8. Booth flags:** a sequential test per booth: 10 visits, up to 30 when unclear.
- **9. District report:** a district whose lower bound is above 5% goes to a full re-check.
- **10. Reproduction pack:** one file anyone runs to get exactly the same number.

Decisions

- **Visits per district:** 1,500 (about ±1.8 points).
- **Seed:** public, announced in advance.
- **Closed door:** three visits.
- **Strata:** yes, by reason.

- **Threshold:** on the lower bound, not the estimate.
- **Who are the checkers?** An independent pool, without knowing who would establish it.

What changed since the POC

In the POC	In the spec
Synthetic district with known truth	Real frozen lists from the Commission export
Honest checker assumed	Double visits to estimate and correct checker error
Simple random sample	Neyman-stratified by reason
Interval only	Booth flags and a district threshold

What stays open

The POC showed that if a quarter of wrong deletions are "confirmed" without a visit, the estimate falls from 14.8% to 11% and the interval covers the truth only 1.8% of the time. The statistics do not protect against that. The spec requires an independent checker pool, and does not say who will build it.